

Intention-Aware Cost Formulation

for Human–Robot Task Scheduling

TeamRob Framework — Technical Reference

Covers: travel cost, intention recognition, goal-position
blockage, interference cost, cancellation cost, task ordering,
and the two-level reselection decision rule.

Applies uniformly to both Mesa and ROS/continuous backends.

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1 Notation and Setup

1.1 Task Structure

The robot is assigned an unordered task set $\mathcal{T}_R = \{\tau_1, \tau_2, \dots, \tau_N\}$. Each task τ_i represents one DELIVER_ITEM instance and decomposes via the HTN decomposition tree as:

$$\text{DELIVER_ITEM}(item_i) \rightarrow \underbrace{\text{move_to}(s_i)}_{\text{STEP*}} \rightarrow \underbrace{\text{pick_up}(item_i)}_{\text{GRASP}} \rightarrow \underbrace{\text{move_to}(g^d)}_{\text{STEP*}} \rightarrow \underbrace{\text{place}(item_i, g^d)}_{\text{RELEASE}}$$

Each task has exactly two *goal positions* — the only positions where human presence causes a waiting cost (PRIEST handles all other co-occupancy through dynamic path planning):

$$g_i^s = \text{position of shelf}_i \quad (\text{pickup goal, from env_layout.json}) \quad (1)$$

$$g_i^d = \text{position of kitting table} \quad (\text{delivery goal, shared by all tasks}) \quad (2)$$

1.2 Robot State

At any cognitive tick, the robot’s state is the tuple:

$$\mathbf{x}_t = (p_t^R, h_t, \tau_t^{\text{cur}}, k_t)$$

where $p_t^R \in \mathbb{R}^2$ is position in metres, $h_t \in \{0, 1\}$ is the **holding** flag (read from `WorldState.predicates`), τ_t^{cur} is the task currently executing, and $k_t \in \{0, 1, 2, 3\}$ is the index of the current action within the grounded action list.

1.3 Symbol Table

Symbol	Unit	Meaning
$\mathcal{T}_R$	—	Robot assigned task set
$\mathcal{H}$	—	Human intention hypothesis set (assigned + foreseeable)
$b_t(h)$	—	Belief probability for hypothesis h at time t
μ_t	—	Observed human micro-action at time t
z_t^H	—	Human’s current zone (from <code>in_zone</code> predicate)
$\omega(h, z)$	$[0, 1]$	Context weight for hypothesis h in zone z
ε	—	Likelihood floor (prevents zero probability; ≈ 0.05)
λ	s^{-1}	Belief decay rate (<code>costs.yaml</code>)
v_R	m/s	Robot speed (<code>costs.yaml</code>)
$\alpha(z_a, z_b)$	—	Geometry correction factor for zone pair
β	—	Global human replanning overhead factor
r_{block}	m (or cells)	Goal blocking radius
θ_{conflict}	$[0, 1]$	Level 1 conflict trigger threshold
$T_{\text{pick}}, T_{\text{place}}$	s	Fixed action durations
c_{oh}	s	Cancellation overhead (not holding)
g_i^s, g_i^d	m	Shelf and delivery goal positions for task τ_i
W_i^s, W_i^d	s	Time windows robot occupies goal positions
σ	—	A permutation (ordering) of remaining tasks
σ^*	—	Optimal ordering

2 Travel Time

Definition 1 (Travel Time). For two positions $p_a, p_b \in \mathbb{R}^2$ in zones z_a, z_b respectively:

$$C_{\text{travel}}(p_a, p_b, z_a, z_b) = \frac{\|p_b - p_a\|_2}{v_R} \cdot \alpha(z_a, z_b) \cdot \beta$$

Euclidean term $\|p_b - p_a\|_2/v_R$. Straight-line travel time assuming constant speed v_R . This is the lower bound on actual travel time.

Geometry correction $\alpha(z_a, z_b) \geq 1$. Accounts for PRIEST trajectory curvature around fixed obstacles. Calibrated *offline* from empty-environment runs: record actual elapsed time vs. straight-line estimate for each zone pair, compute the ratio, and store in `costs.yaml`. Default $\alpha = 1.2$ when the zone pair is not found.

Human correction $\beta \geq 1$. A single global scalar capturing the additional PRIEST replanning overhead when a human is present and moving. Calibrated from runs *with* a human. Applied uniformly to all travel estimates.

Mesa equivalence. Replace positions with grid cells and speed with cells-per-step:

$$C_{\text{travel}}^{\text{Mesa}} = \left\lceil \|c_b - c_a\|_2 / v_{\text{cells}} \right\rceil \cdot \Delta t_{\text{step}} \cdot \alpha(z_a, z_b) \cdot \beta$$

where c_a, c_b are grid cell coordinates, v_{cells} is cells per step, and Δt_{step} is seconds per step. In Mesa, $\alpha = 1.0$ is appropriate (paths are exact in discrete grids without obstacles).

3 Arrival Time Windows

Definition 2 (Arrival Times). For task τ_i starting from position p at time t_0 :

$$\begin{aligned} \eta_i^s(p, t_0) &= t_0 + C_{\text{travel}}(p, g_i^s, z_p, z_i^s) \\ \eta_i^d(p, t_0) &= \eta_i^s + T_{\text{pick}} + C_{\text{travel}}(g_i^s, g_i^d, z_i^s, z_i^d) \end{aligned}$$

These define two *goal-occupancy windows*:

$$\begin{aligned} W_i^s &= [\eta_i^s, \eta_i^s + T_{\text{pick}}] \quad (\text{robot at shelf}_i) \\ W_i^d &= [\eta_i^d, \eta_i^d + T_{\text{place}}] \quad (\text{robot at kitting table}) \end{aligned}$$

The time windows W_i^s and W_i^d are the only intervals during which human presence at g_i^s or g_i^d respectively causes a real waiting cost. Outside these windows, PRIEST navigates freely regardless of human position.

4 Intention Recognition and Belief State

4.1 Bayesian Belief Update

The robot maintains a belief distribution b_t over the human’s intention hypothesis set $\mathcal{H} = \{h_1, \dots, h_M\}$, which includes both assigned tasks and foreseeable tasks (e.g. COFFEE_BREAK, A/C_ACTIVATION).

Definition 3 (Bayesian Belief Update). After observing micro-action μ_t with human in zone z_t^H :

$$\tilde{b}_t(h) = \underbrace{P(\mu_t \mid h, \text{progress}_t(h))}_{\text{likelihood}} \cdot \underbrace{b_{t-1}(h)}_{\text{prior}} \cdot \underbrace{\omega(h, z_t^H)}_{\text{context weight}}$$

$$b_t(h) = \frac{\tilde{b}_t(h)}{\sum_{h' \in \mathcal{H}} \tilde{b}_t(h')}$$

4.2 Likelihood

The likelihood asks: if the human were executing hypothesis h , how expected is the observed micro-action μ_t ? The decomposition tree of h defines a sequence of expected micro-actions. A per-hypothesis *progress pointer* tracks how far through the tree the human has advanced based on observation history.

$$P(\mu_t \mid h, \text{progress}_t(h)) = \begin{cases} 1.0 & \text{if } \mu_t \text{ is the expected next micro-action under } h \\ \varepsilon & \text{if } \mu_t \text{ is possible but unexpected under } h \\ \varepsilon & \text{if } \mu_t \text{ is inconsistent with } h \end{cases}$$

The floor $\varepsilon \approx 0.05$ prevents any hypothesis reaching zero permanently, reflecting that humans occasionally deviate slightly from expected behaviour.

Observable micro-actions are defined in the domain action schemas: STEP*, GRASP, RELEASE, STAND*.

4.3 Context Weight

The context weight uses the `in_zone` predicate from `WorldState` — the only predicate used for zone-level reasoning, strictly separate from the `at` predicate used for completion checking.

$$\omega(h, z_t^H) = \begin{cases} 1.0 & \text{if } z_t^H \text{ is the expected zone under } h \text{ at current progress stage} \\ 0.5 & \text{if } z_t^H \in \text{zones}(h) \text{ but not at this stage} \\ 0.1 & \text{if } z_t^H \notin \text{zones}(h) \text{ at all} \end{cases}$$

4.4 Belief Decay over Projection Horizon

When projecting the belief forward by Δt seconds to estimate future human positions, uncertainty grows. The belief is propagated as:

Definition 4 (Belief Propagation).

$$b_{t+\Delta t}(h) = b_t(h) \cdot e^{-\lambda \Delta t} + \frac{1 - e^{-\lambda \Delta t}}{|\mathcal{H}|}$$

As $\Delta t \rightarrow \infty$, the belief returns to the uniform distribution, reflecting that long-range predictions are increasingly unreliable. λ is stored in `costs.yaml` as `belief_decay_per_second`.

5 Goal Position Blockage Probability

Key correction from zone-level reasoning. PRIEST handles general co-occupancy between human and robot dynamically. A waiting cost arises *only* when the human stands at or immediately adjacent to the robot’s current goal position. Zone-level co-occupancy does *not* create a cost in general.

Definition 5 (Goal Position Blockage). For goal position g , time window $[t_1, t_2]$, and belief b_t :

$$P_{\text{block}}(g, t_1, t_2, b_t) = \sum_{h \in \mathcal{H}} b_t(h) \cdot \mathbf{1} \left[\|g_h^{\text{pred}}(\bar{t}) - g\|_2 \leq r_{\text{block}} \right] \cdot \mathbf{1} \left[[\eta_h^{\text{pred}}, \eta_h^{\text{pred}} + d_h(g)] \cap [t_1, t_2] \neq \emptyset \right]$$

where $\bar{t} = (t_1 + t_2)/2$.

Terms explained.

- $g_h^{\text{pred}}(\bar{t})$: the position hypothesis h predicts the human to visit near time $\bar{t}$, read from h ’s decomposition tree.
- r_{block} : blocking radius. In continuous (ROS): $r_{\text{block}} \approx \max(a_{\text{obs}}, b_{\text{obs}})$ from PRIEST obstacle parameters (`a_obs_1`, `b_obs_1` in planner config). In Mesa: $r_{\text{block}} = 1$ cell (Chebyshev distance).
- η_h^{pred} : predicted time the human arrives at g_h^{pred} , computed using the same η formula applied to the human’s task under h .
- $d_h(g)$: expected human dwell time at goal g under h . If g is a shelf: $d_h(g) = T_{\text{pick}}$. If g is the kitting table: $d_h(g) = T_{\text{place}}$.

Spatial overlap condition. For Mesa, replace the Euclidean distance with Chebyshev distance:

$$\|g_h^{\text{pred}} - g\|_{\infty} = \max(|\Delta_{\text{row}}|, |\Delta_{\text{col}}|) \leq 1$$

Non-zero condition. $P_{\text{block}} > 0$ requires *both* spatial overlap (human predicted at the same goal) and temporal overlap (windows intersect). In the kitting domain this is most likely at the kitting table, where all tasks share the same delivery goal and both human and robot converge.

6 Interference Cost

Definition 6 (Interference Cost). For task τ_i starting from position p at time t_0 :

$$C_{\text{int}}(\tau_i, p, b_t, t_0) = \underbrace{P_{\text{block}}(g_i^s, W_i^s, b_t) \cdot \mathbb{E}[D(g_i^s, b_t)]}_{\text{shelf blockage}} + \underbrace{P_{\text{block}}(g_i^d, W_i^d, b_t) \cdot \mathbb{E}[D(g_i^d, b_t)]}_{\text{delivery blockage}} + \underbrace{C_{\text{bot}}(\tau_i, p, b_t, t_0)}_{\text{narrow passages}}$$

6.1 Expected Dwell Time

$$\mathbb{E}[D(g, b_t)] = \sum_{h \in \mathcal{H}} b_t(h) \cdot d_h(g)$$

where $d_h(g) = T_{\text{pick}}$ if g is a shelf under h , $d_h(g) = T_{\text{place}}$ if g is the kitting table under h , and $d_h(g) = 0$ otherwise.

6.2 Bottleneck Cost

Narrow passages are the *only* case where zone-level co-occupancy reasoning is appropriate, because blockage there depends on corridor width, not a specific goal point. Bottleneck positions $\mathcal{B}$ are identified once from the layout topology.

$$C_{\text{bot}}(\tau_i, p, b_t, t_0) = \sum_{b \in \mathcal{B}} P_{\text{block}}(p_b, [t_b, t_b + T_{\text{transit}}], b_t) \cdot \Delta_b$$

where $t_b = t_0 + C_{\text{travel}}(p, p_b, \dots)$ is when the robot reaches bottleneck b , and Δ_b is its precomputed expected delay.

7 Single Task Cost

Definition 7 (Single Task Cost). For task τ_i starting from position p at time t_0 :

$$C_{\text{task}}(\tau_i, p, b_t, t_0) = \underbrace{C_{\text{travel}}(p, g_i^s) + T_{\text{pick}} + C_{\text{travel}}(g_i^s, g_i^d) + T_{\text{place}}}_{C_{\text{base}}(\tau_i, p) \text{ — geometry only}} + \underbrace{C_{\text{int}}(\tau_i, p, b_t, t_0)}_{\text{human-induced delay}}$$

The task finish time is:

$$t_i^{\text{finish}} = t_0 + C_{\text{task}}(\tau_i, p, b_t, t_0)$$

Graceful fallback. When belief is uniform (IR not yet active), $P_{\text{block}} = 0$ for all tasks and the cost reduces to C_{base} — equivalent to nearest-shelf-first ordering. This is the correct baseline behaviour.

8 Task Ordering Cost

Definition 8 (Ordering Cost). For ordering $\sigma = (\tau_{\sigma(1)}, \dots, \tau_{\sigma(N)})$, tasks are chained sequentially:

$$C_{\text{ord}}(\sigma, p_0, b_{t_0}) = \sum_{k=1}^N C_{\text{task}}(\tau_{\sigma(k)}, p_{k-1}, b_{t_{k-1}}, t_{k-1})$$

where at each step:

$$\begin{aligned} p_k &= g_{\sigma(k)}^d \quad (\text{robot ends at delivery position of task } k) \\ t_k &= t_{k-1}^{\text{finish}} \\ b_{t_k} &= \text{propagate}(b_{t_{k-1}}, t_k - t_{k-1}, \lambda) \end{aligned}$$

Optimal ordering.

$$\sigma^* = \arg \min_{\sigma \in \text{Perm}(\mathcal{T}_R^{\text{rem}})} C_{\text{ord}}(\sigma, p_0, b_{t_0})$$

- $|\mathcal{T}_R^{\text{rem}}| \leq 6$: enumerate all $N!$ permutations (at most 720; each evaluation is arithmetic, so real-time safe).
- $|\mathcal{T}_R^{\text{rem}}| > 6$: greedy nearest-cost insertion ($O(N^2)$, good approximation for moderate N).

9 Cancellation Cost

Definition 9 (Cancellation Cost). The cost of abandoning the current task mid-execution:

$$C_{\text{cancel}}(\mathbf{x}_t) = \begin{cases} c_{\text{oh}} & h_t = 0 \quad (\text{not holding — just stop}) \\ C_{\text{travel}}(p_t^R, g_{\tau^{\text{cur}}}^s) + T_{\text{place}} & h_t = 1 \quad (\text{holding — must return item to shelf}) \end{cases}$$

The **discontinuity** at $h_t = 0 \rightarrow 1$ is the essential property of this cost. It occurs at the exact moment `pick_up` completes and `holding(robot, item)` enters `WorldState.predicates`. After this point, abandonment always incurs a return-to-shelf penalty, creating a natural *commit boundary* at pickup.

10 Two-Level Decision Rule

10.1 Level 1 — Current Task Interference Check

Run at every cognitive tick.

Step 1. Identify the current action’s goal position g_{cur} (either g_i^s or g_i^d , depending on action index k_t) and compute:

$$P_{\text{conflict}} = P_{\text{block}}(g_{\text{cur}}, [0, T_{\text{rem}}], b_t)$$

where T_{rem} is the estimated remaining time to reach g_{cur} .

Step 2. If $P_{\text{conflict}} \geq \theta_{\text{conflict}}$:

Definition 10 (Level 1 Decision).

$$\text{Decision} = \begin{cases} \text{WAIT} & h_t = 1 \quad (\text{holding} - \text{cannot abandon}) \\ \text{WAIT} & C_{\text{wait}} + C_{\text{ord}}([\tau^{\text{cur}}] \cup \sigma_{\text{rest}}^*) \leq C_{\text{cancel}} + C_{\text{ord}}(\sigma_{\text{all}}^*) \\ \text{RESELECT} & C_{\text{cancel}} + C_{\text{ord}}(\sigma_{\text{all}}^*) < C_{\text{wait}} + C_{\text{ord}}([\tau^{\text{cur}}] \cup \sigma_{\text{rest}}^*) \end{cases}$$

where $C_{\text{wait}} = \mathbb{E}[D(g_{\text{cur}}, b_t)]$.

If $P_{\text{conflict}} < \theta_{\text{conflict}}$: continue current task without interruption and proceed to Level 2.

10.2 Level 2 — Future Task Ordering

Runs when (a) the current task completes, or (b) Level 1 finds no conflict.

Definition 11 (Level 2 Ordering).

$$\sigma_{\text{next}}^* = \arg \min_{\sigma} C_{\text{ord}}(\sigma, g_{\tau^{\text{cur}}}^d, \text{propagate}(b_t, T_{\text{rem}}, \lambda))$$

The belief is propagated forward by T_{rem} before evaluating future ordering costs. This means inference estimates for future tasks use an appropriately more uncertain belief.

11 Backend Equivalence: Mesa vs. ROS

All cost functions in `shared/costs.py` operate in **seconds**. Both backends convert their native units through layout adapters before calling shared functions. The decision logic in `shared/meta_planner.py` is *identical* for both backends.

Quantity	Mesa	Continuous (ROS)
Position	Grid cell (r, c) , converted to metres	(x, y) metres directly
Distance	$\ c_b - c_a\ _2 \times c_{\text{cell}}$	$\ p_b - p_a\ _2$
r_{block}	1 cell (Chebyshev distance)	$\max(a_{\text{obs}}, b_{\text{obs}})$ metres
v_R	$c_{\text{cell}} / \Delta t_{\text{step}}$ m/s	Constant from <code>costs.yaml</code>
α, β	1.0 (exact paths) or calibrated	Calibrated from runs
$T_{\text{pick}}, T_{\text{place}}$	Steps $\times \Delta t_{\text{step}}$	Seconds
$b_t(h)$	Identical Bayesian update	Identical Bayesian update
at predicate	Within PROXIMITY_THRESHOLD cells	Within PROXIMITY_THRESHOLD metres
in_zone predicate	Cell within zone bounds	Position within zone bounds
Cost unit	Seconds (after conversion)	Seconds

12 Cost Parameter Summary

All tunable parameters are stored in `costs.yaml`. No hard-coded constants appear in `shared/costs.py`.

Parameter key	Symbol	How to calibrate
<code>robot.speed.ms</code>	v_R	Robot specification
<code>actions.T_pick</code>	T_{pick}	Timed trials
<code>actions.T_place</code>	T_{place}	Timed trials
<code>actions.cancel_overhead</code>	c_{oh}	Estimated stop + reorientation
<code>priest_correction.zones.*</code>	$\alpha(z_a, z_b)$	Empty-environment runs per zone pair
<code>human_correction</code>	β	Runs with human present
<code>blocking.continuous_radius_m</code>	r_{block}	From PRIEST <code>a_obs_1</code>
<code>blocking.mesa.radius_cells</code>	r_{block}	Fixed at 1
<code>interference.ignore_threshold</code>	—	Minimum P_{block} to count
<code>interference.belief_decay_per_second</code>	λ	Cross-validated from IR data
<code>reselect.conflict_threshold</code>	θ_{conflict}	Empirical tuning